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Teorema de parada opcional

Teorema 1 (de parada opcional). *Sea* $(X_n)_n$ *un proceso estocástico y* $S \leqslant T$ *tiempos de parada respecto a la filtración* $(\mathcal{F}_n)_n$ *tales que* $T < \infty$ *c.s., entonces*

(1) *Si* $(X_n)_n$ *es una martingala respecto a* $(\mathcal{F}_n)_n \implies \mathbb{E}\left[X_T \mid \mathcal{F}_S\right] = X_S$ *c.s.*

(2) *Si* $(X_n)_n$ *es una submartingala respecto a* $(\mathcal{F}_n)_n \implies \mathbb{E}\left[X_T \mid \mathcal{F}_S\right] \geqslant X_S$ *c.s.*

(3) *Si* $(X_n)_n$ *es una supermartingala respecto a* $(\mathcal{F}_n)_n \implies \mathbb{E}\left[X_T \mid \mathcal{F}_S\right] \leqslant X_S$ *c.s.*

donde X_S *está dada por* $\forall \omega \in \Omega : X_S(\omega) = X_{S(\omega)}(\omega)$.

***Demostración*:** Consultar **durrettProbabilityTheoryExamples2019**. ∎

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